Series

Example ① Single stub match. @ $Z_0 = 50\Omega$

$Z_L = 20 + 10j\ \Omega$ (ohms so impedance)

① Normalize value. Impedance so $\div Z_0$

$z = 0.4 + 0.2j$

② Plot on chart. z

③ Move away from the load along the transmission line toward the generator, clockwise round the VSWR circle ③

④ The first option is A on the matching circle the shortest distance along the line then B. so 0.125λ to A & 0.302λ to B

⑤ A = 1+j Ω so +j needs removing so we will add a stub with a value of −j

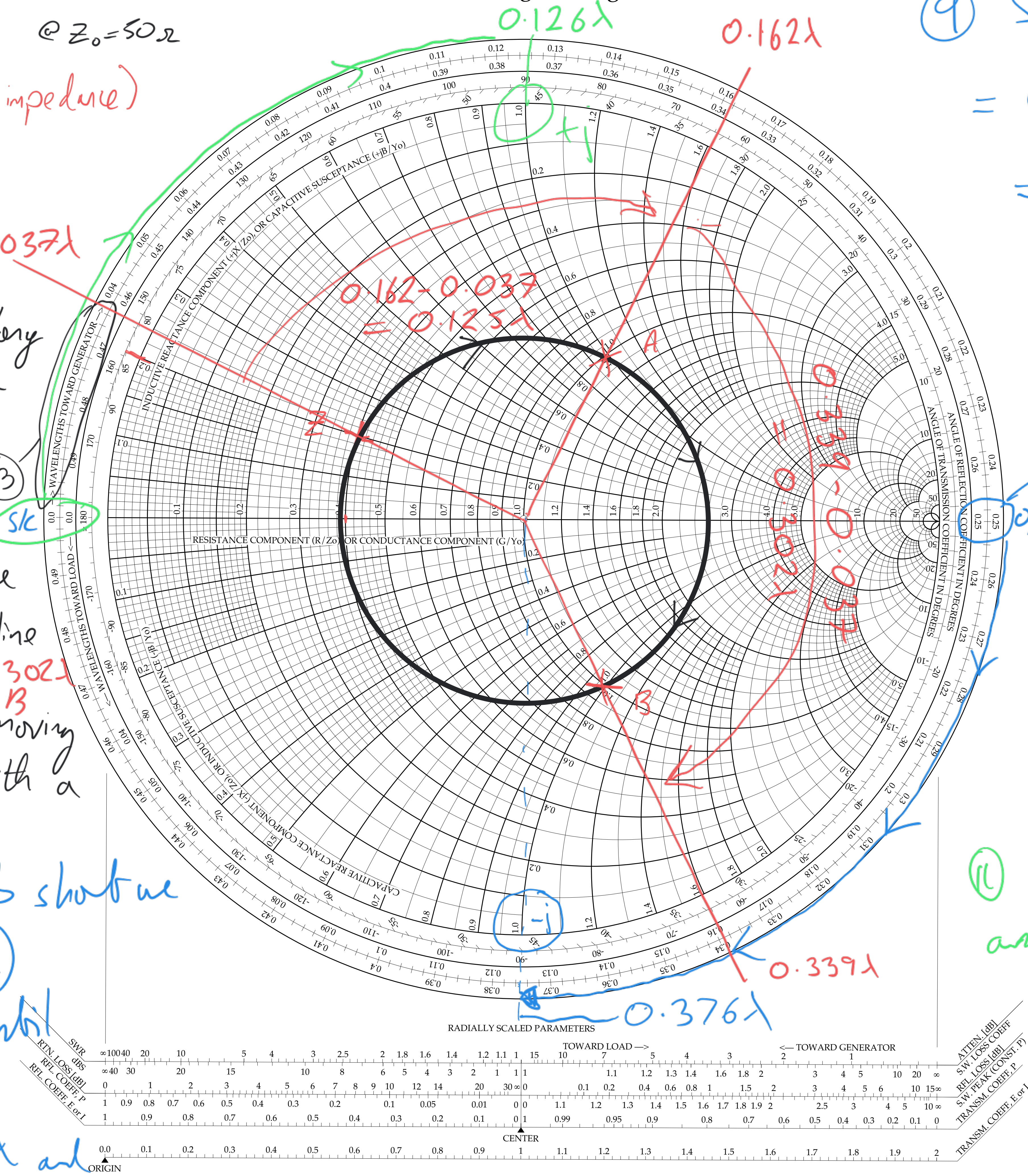
⑥ Add stub. To keep stub short we choose an open cct. (o/c)

⑦ We lengthen the stub until we hit −1j.

⑧ So starting at 0.25λ and rotating toward the generator until −1j at 0.376λ

The Complete Smith Chart

Black Magic Design



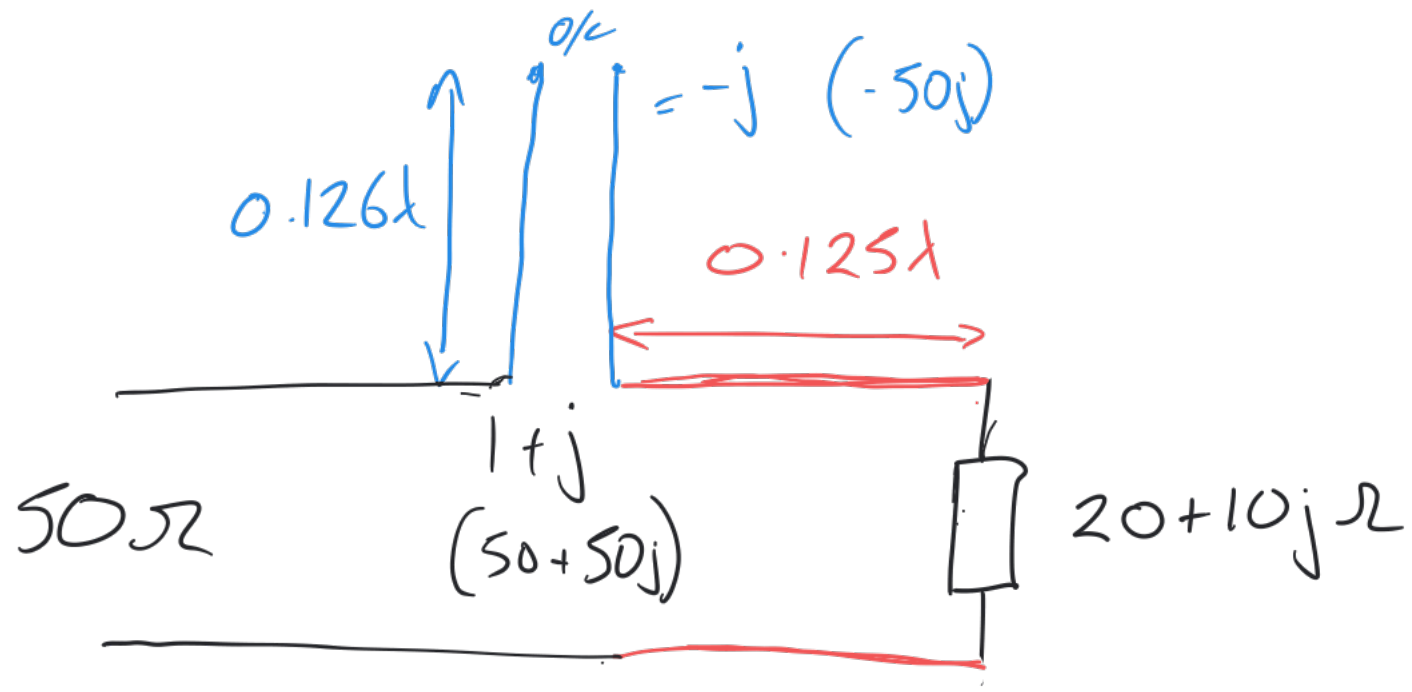
⑨ Stub length = 0.376 − 0.25 = 0.126λ o/c

⑩ Option 2 from B = 1 − j. We need to add +j to get to a match.

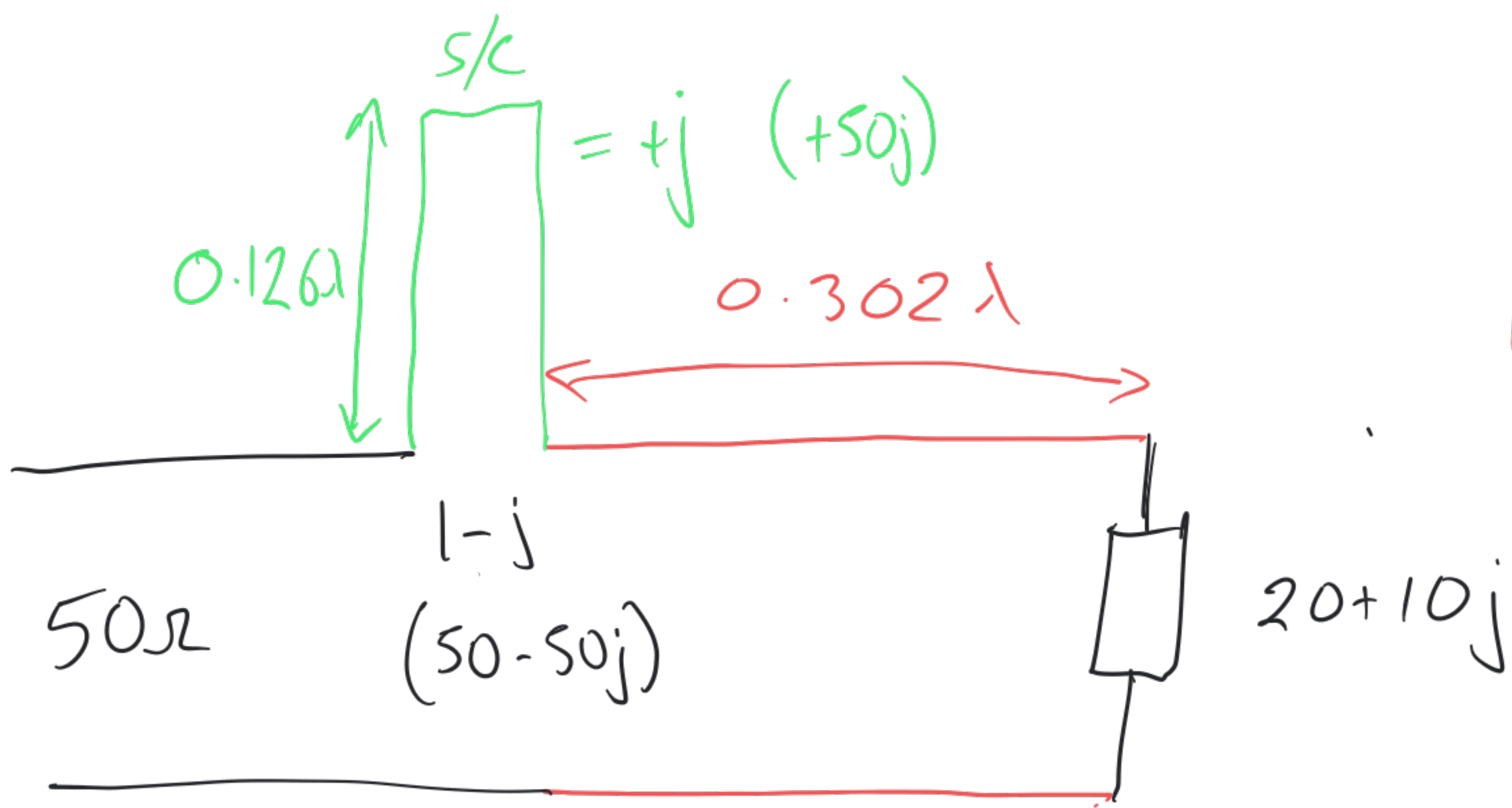
⑪ Choose s/c stub and lengthen until it equals +j

⑫ So from 0λ to 0.126λ

stub length = 0.126λ s/c



A. Rotate 0.125λ towards gen to hit matching circle at 1+j. Add o/c stubs of length 0.126λ giving value of −j.



B. Rotate 0.302λ onto matching circle at 1−j. Add short cct stubs of length 0.126λ to give value of +1j

The Complete Smith Chart

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Parallel single stub.

$Z_L = 100 - 100j\ \Omega$

$(\div 50) \Rightarrow 2 - 2j$

Plot Z. Transpose to Y and admittance – parallel.

Through centre same distance other side.

$y = 0.26 + 0.24j$

Rotate onto matching circle towards gen.

From $0.04 \to 0.18$

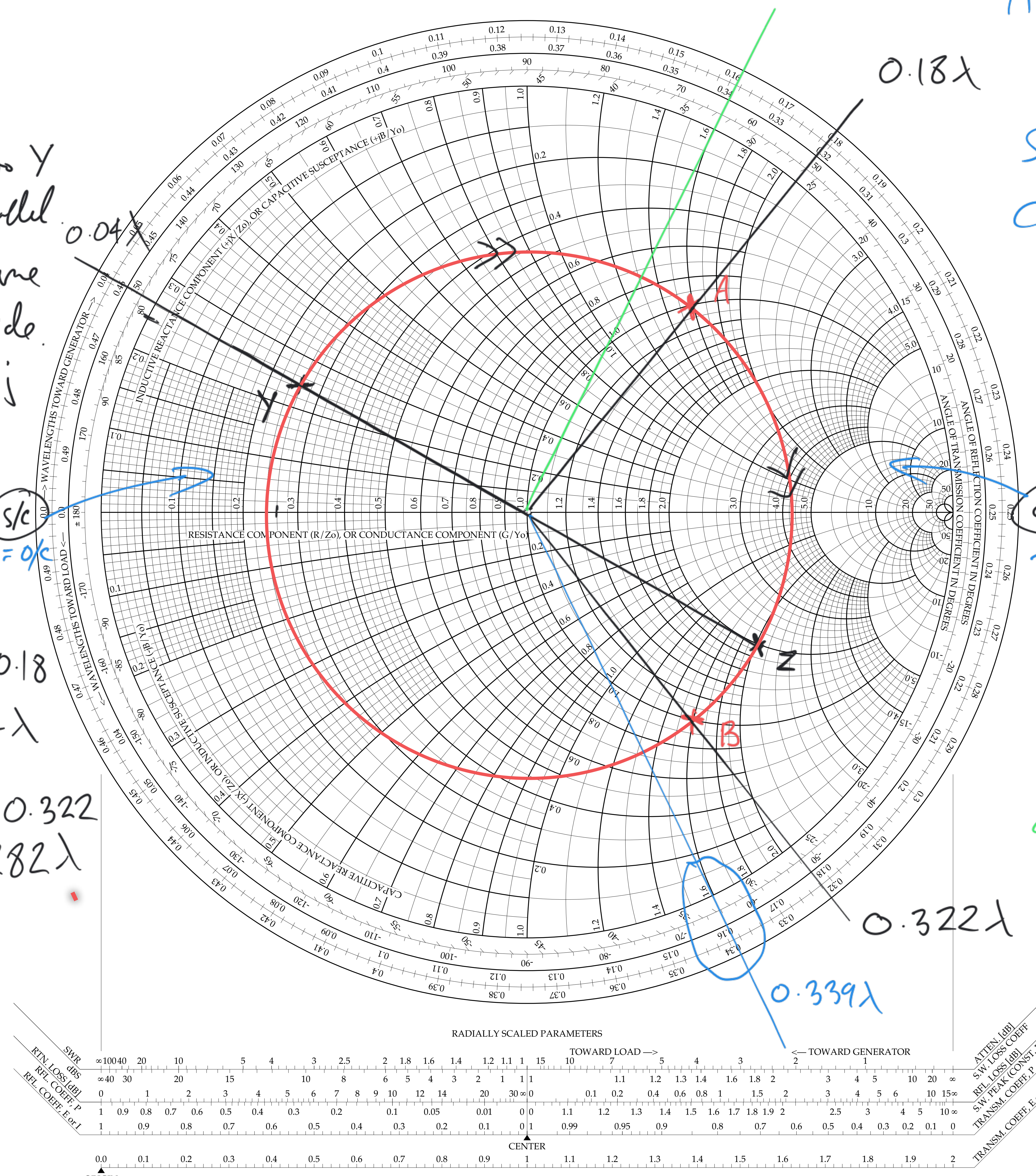
$A = 0.14\lambda$

& $0.04 \to 0.322$

$B = 0.282\lambda$

$A = 1 + 1.6j$

$B = 1 - 1.6j$



Add stub to cancel $+1.6j$ at A.

So stub must $= -1.6j$

Choose s/c length

$= 0.339 - 0.25$

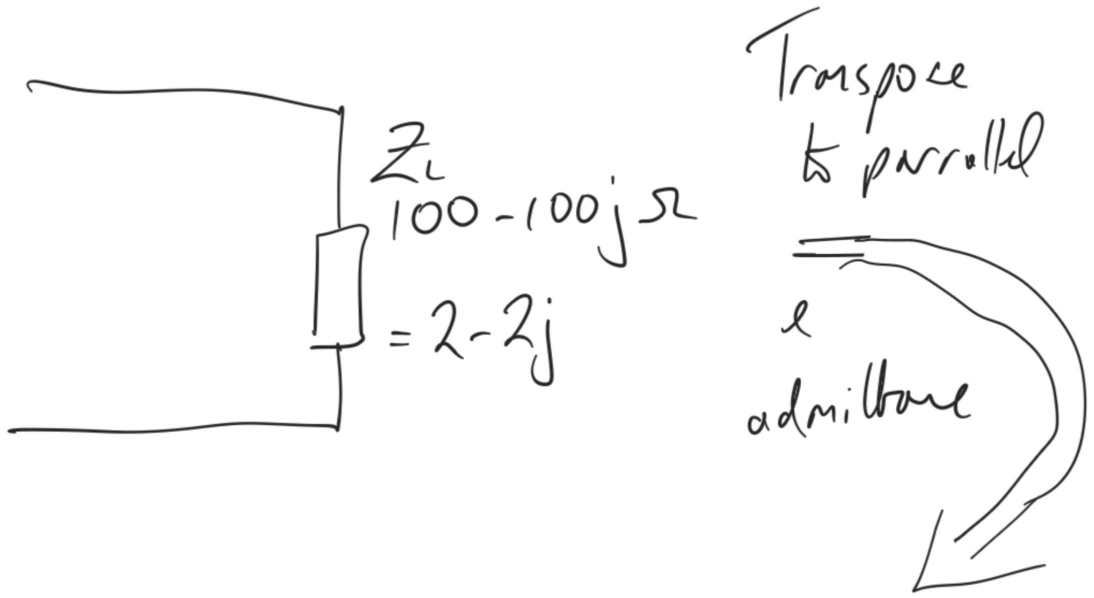
$= 0.089\lambda$

Transpose = s/c

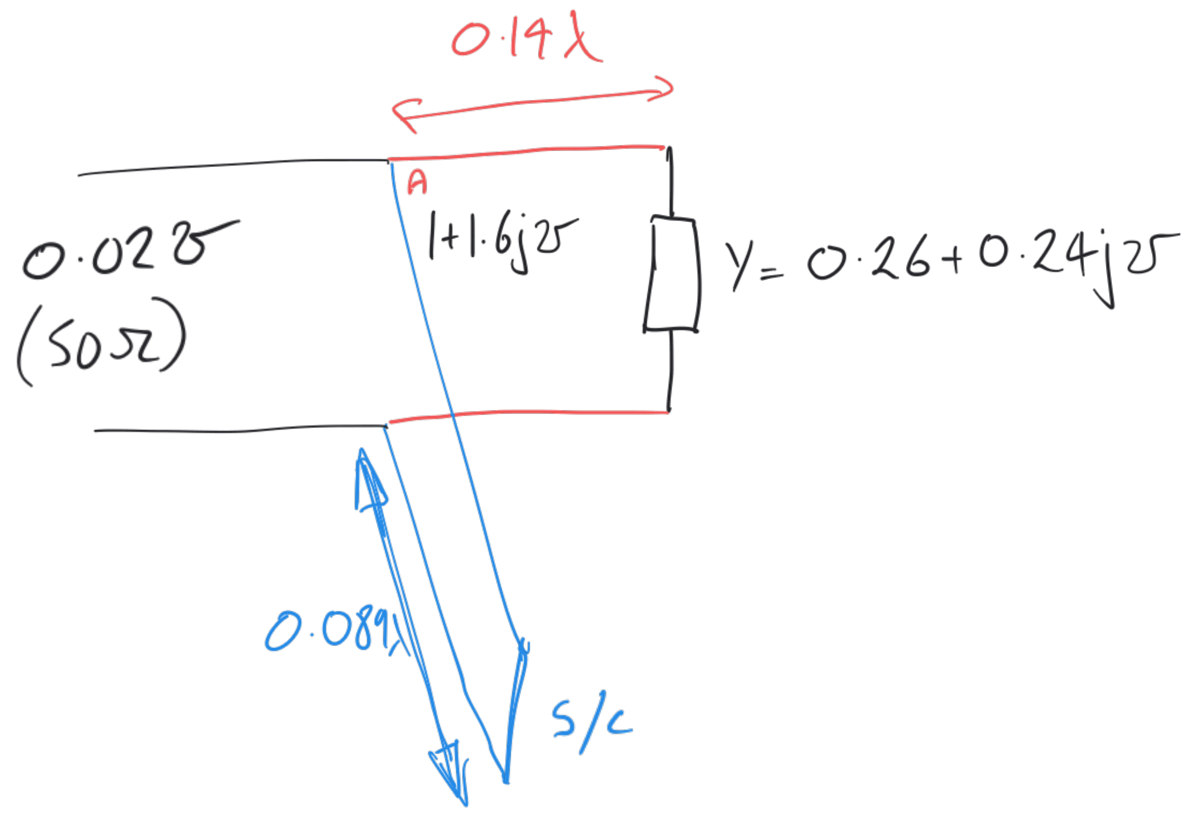
Add stub to add $1.6j$ if option B is chosen.

Choose o/c length

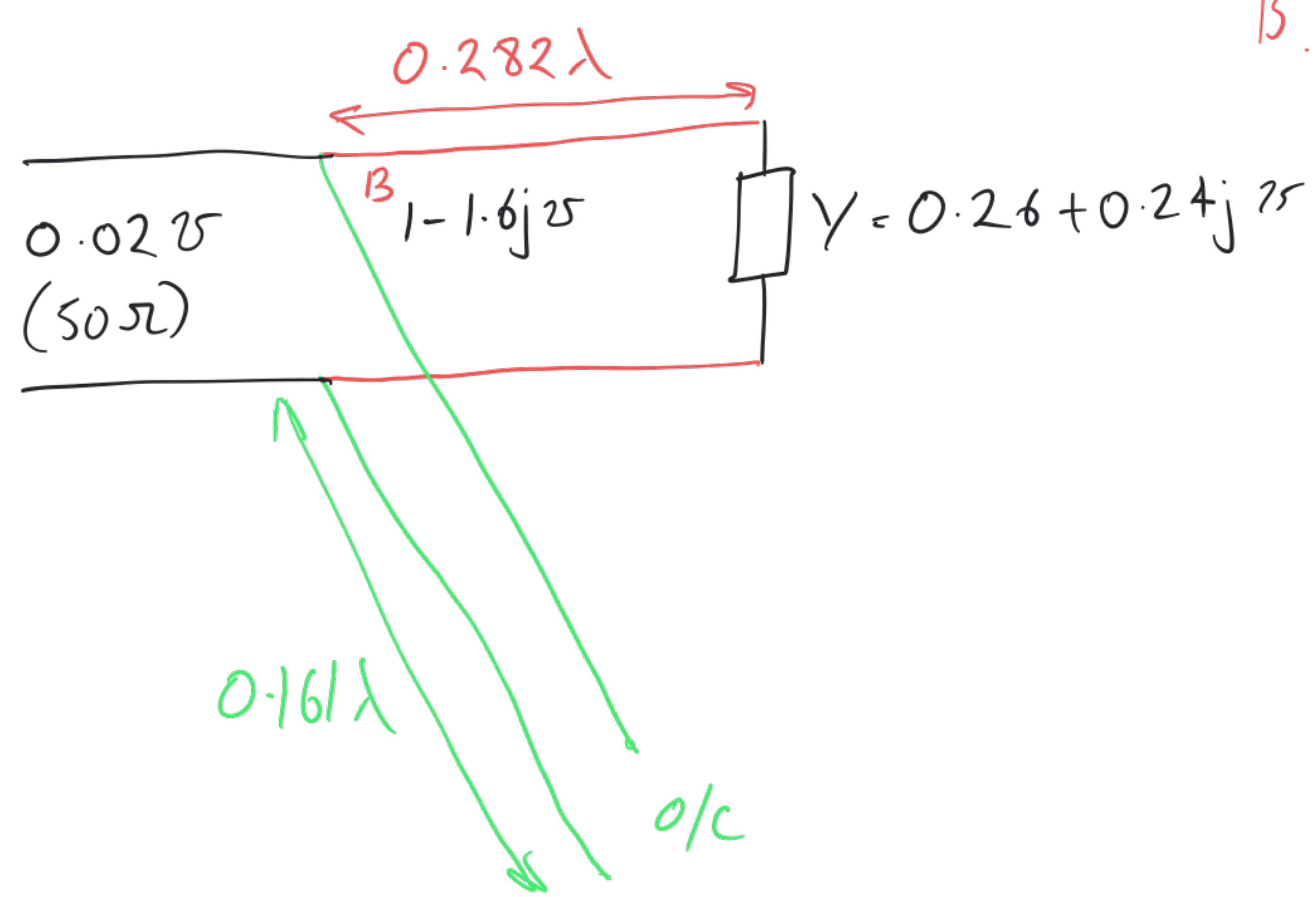
0.161λ



Transpose your impedance to parallel and an admittance.
You would not need to do this if you were given an admittance to start with.
Transpose the o/c e s/c points to admittance.



A. Rotate Y 0.14λ onto matching circle
Add stub to cancel +1.6j
In this case a s/c stub of length 0.089λ
= −1.6j ℧



B. Or.. rotate Y 0.282λ onto pt B on matching circle.
Add stub to cancel −1.6j
In this case an o/c stub, length 0.161λ = +1.6j

The Complete Smith Chart

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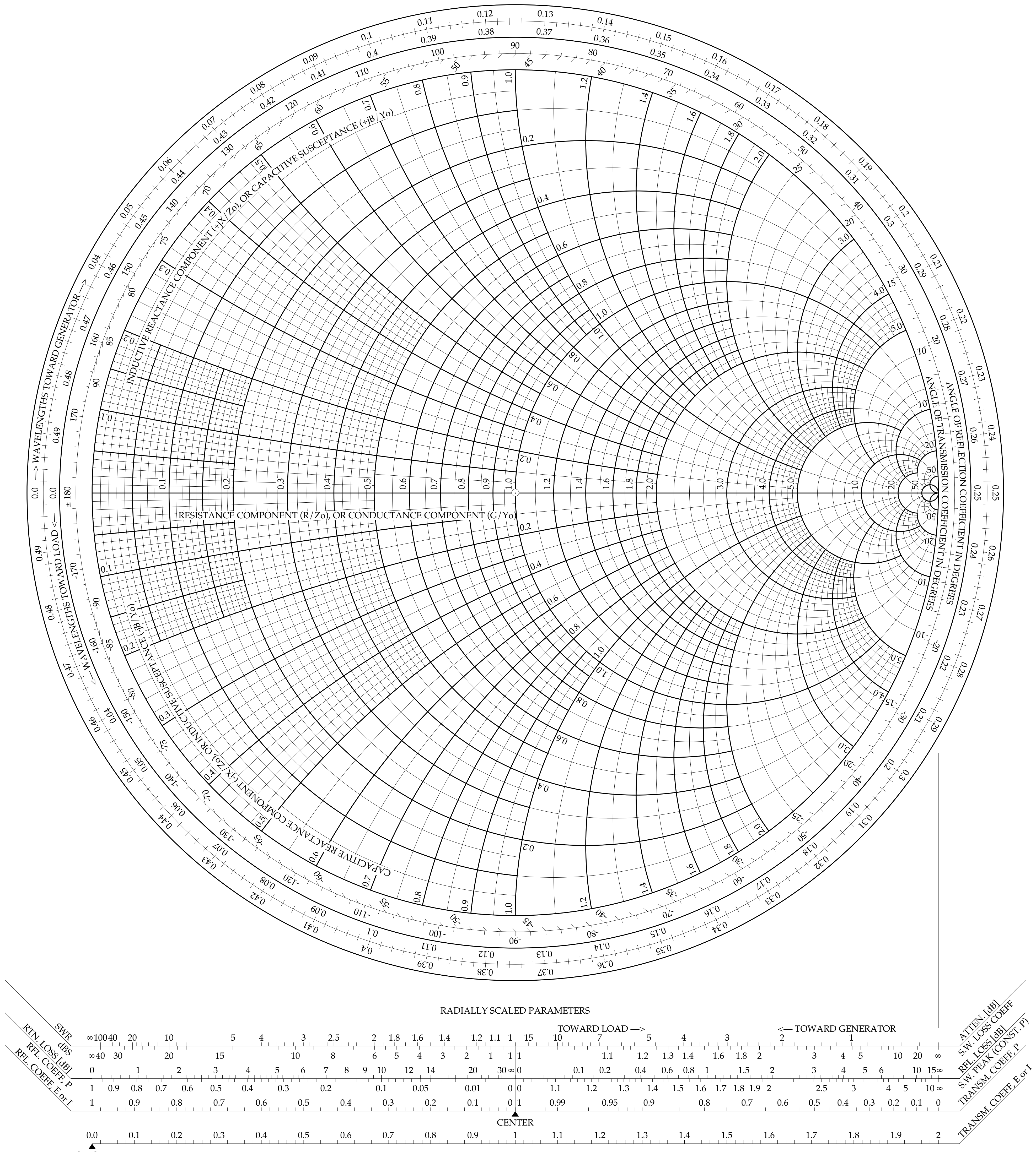

The Complete Smith Chart

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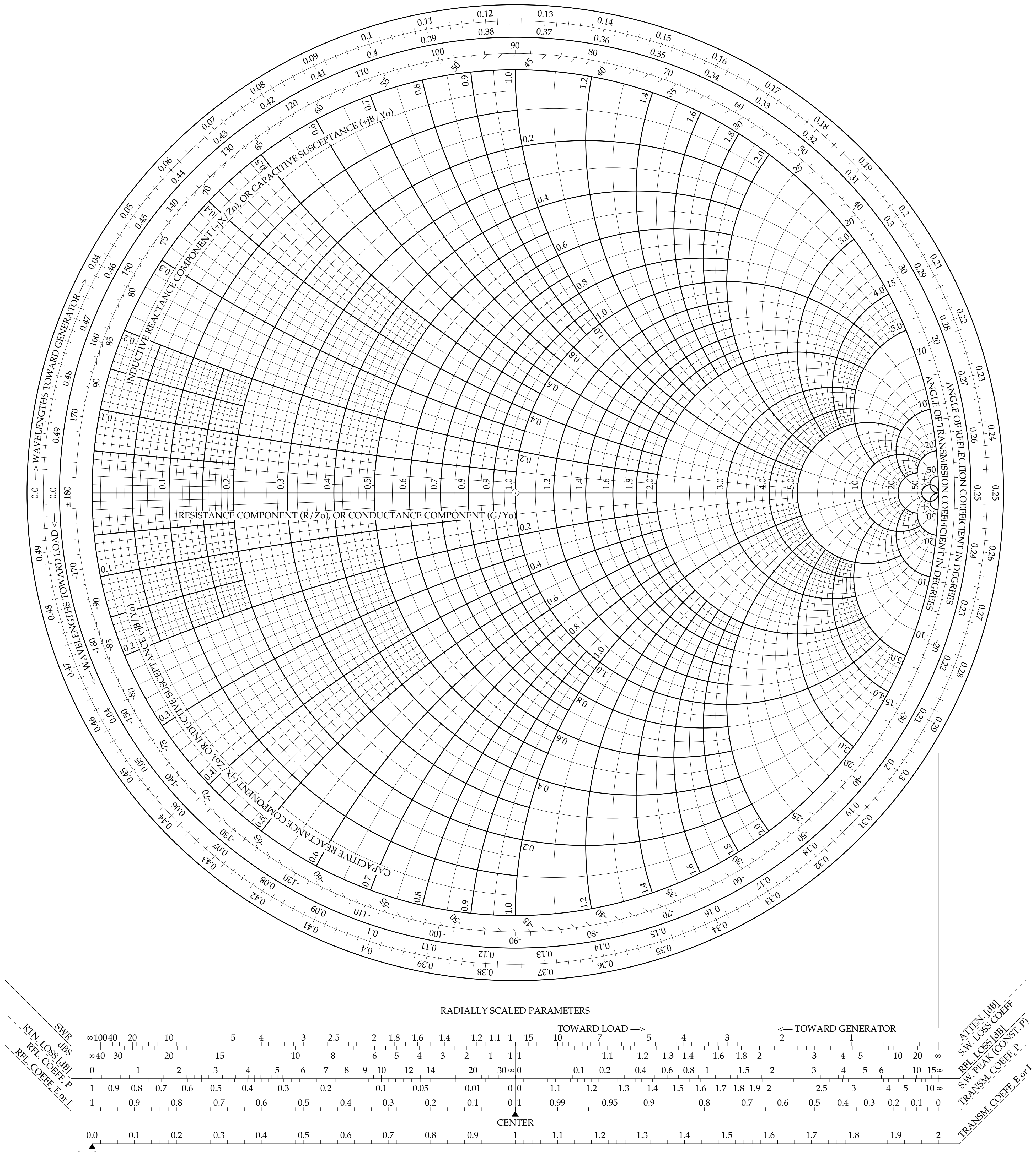

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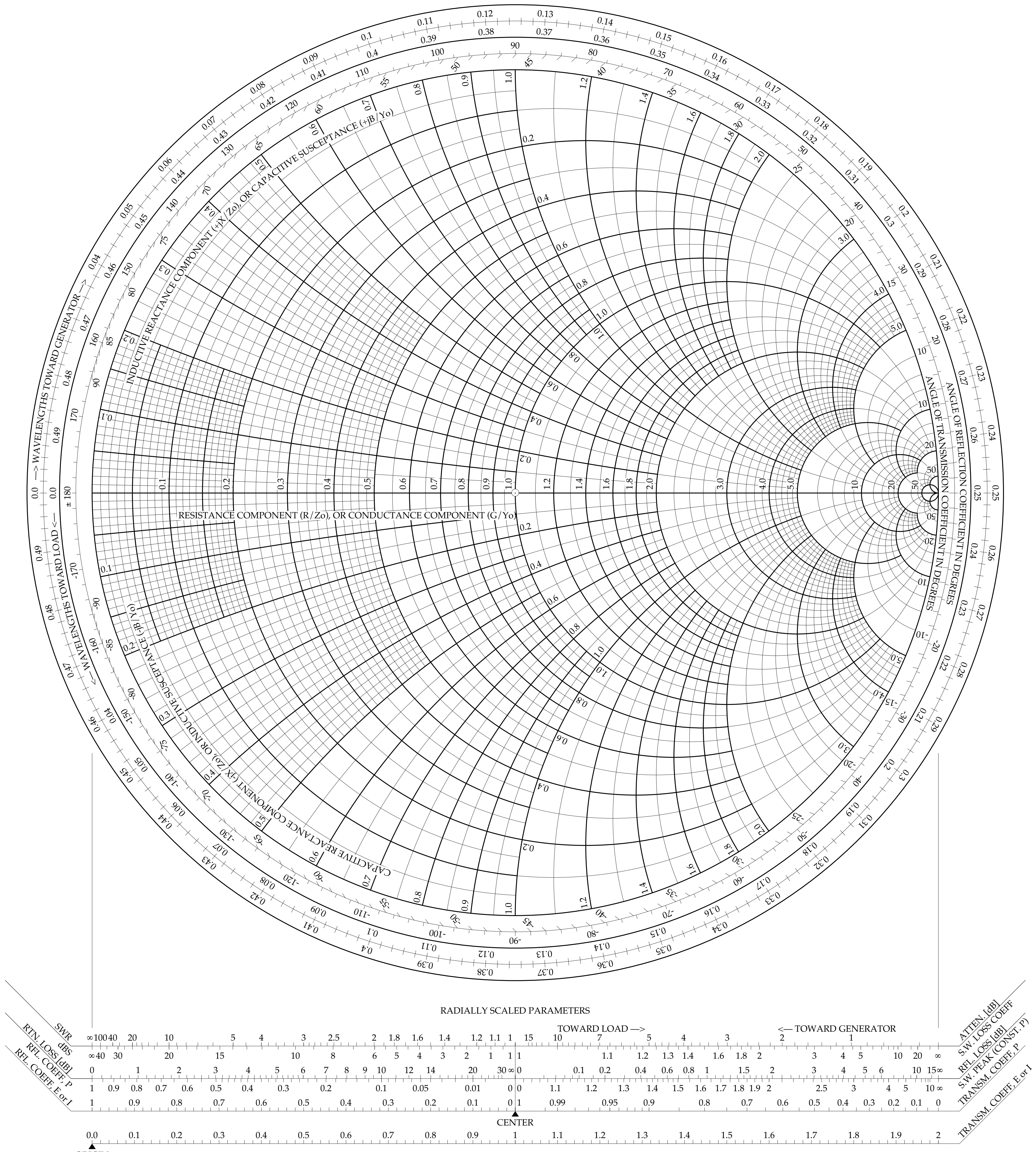

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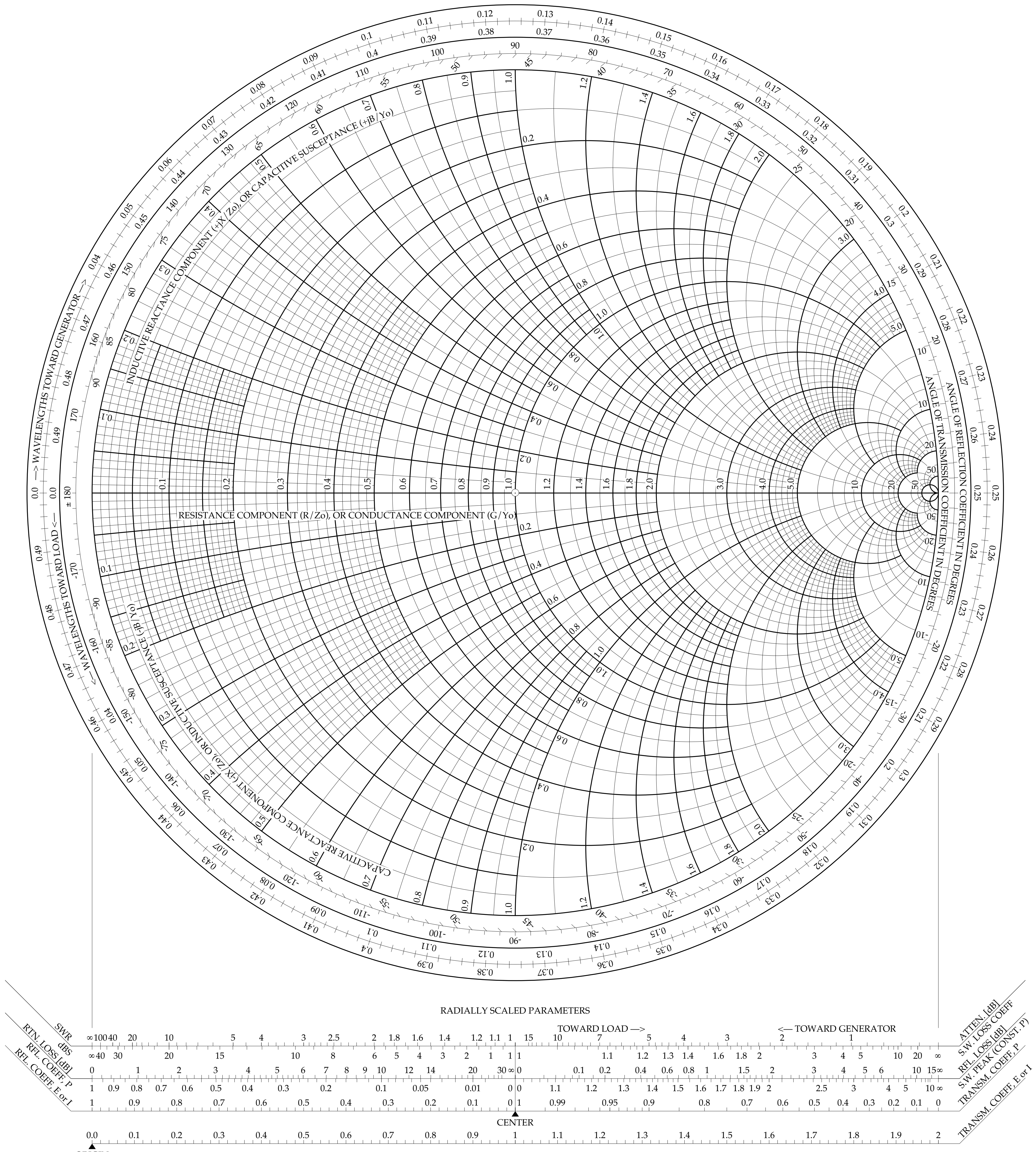

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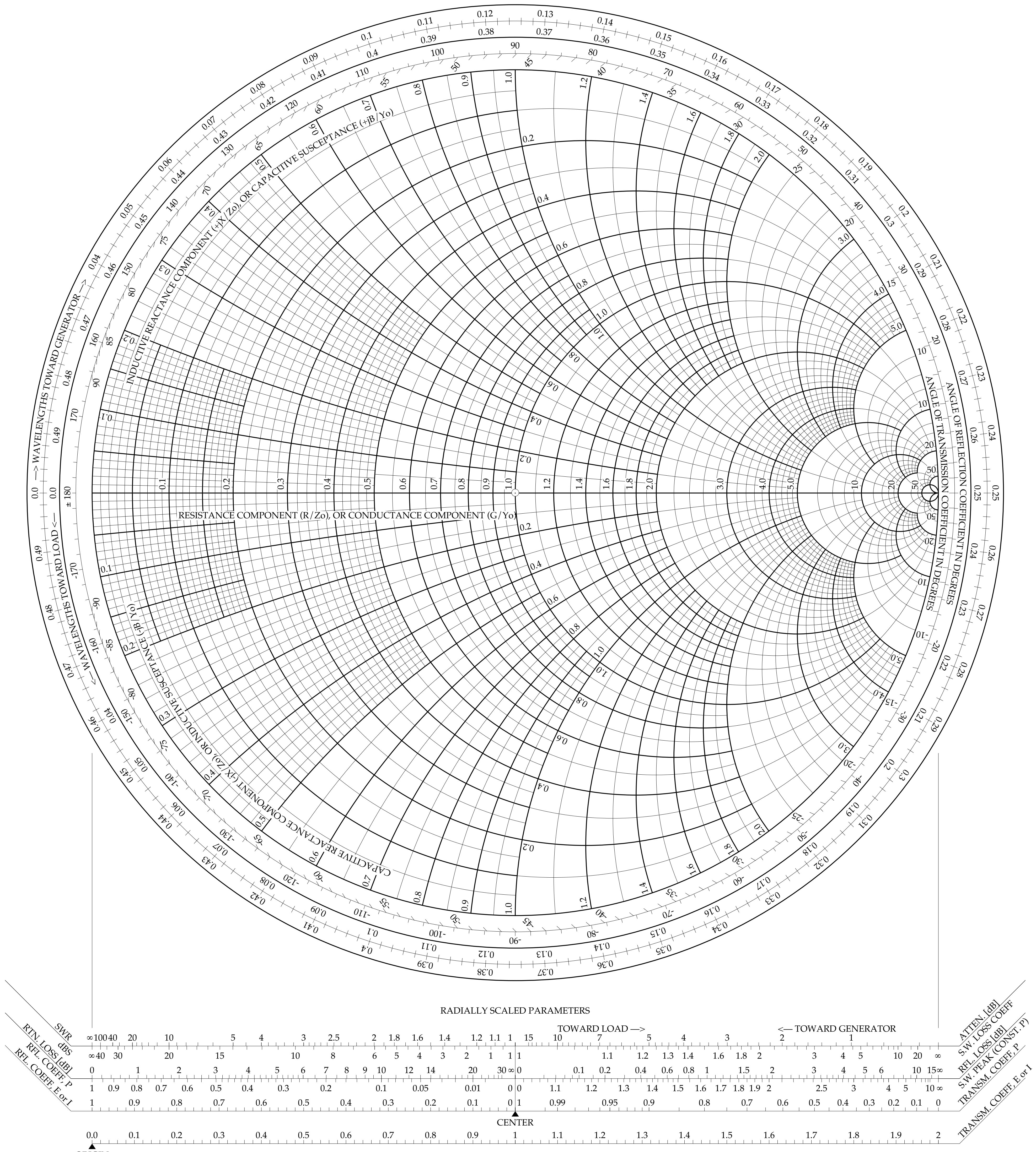

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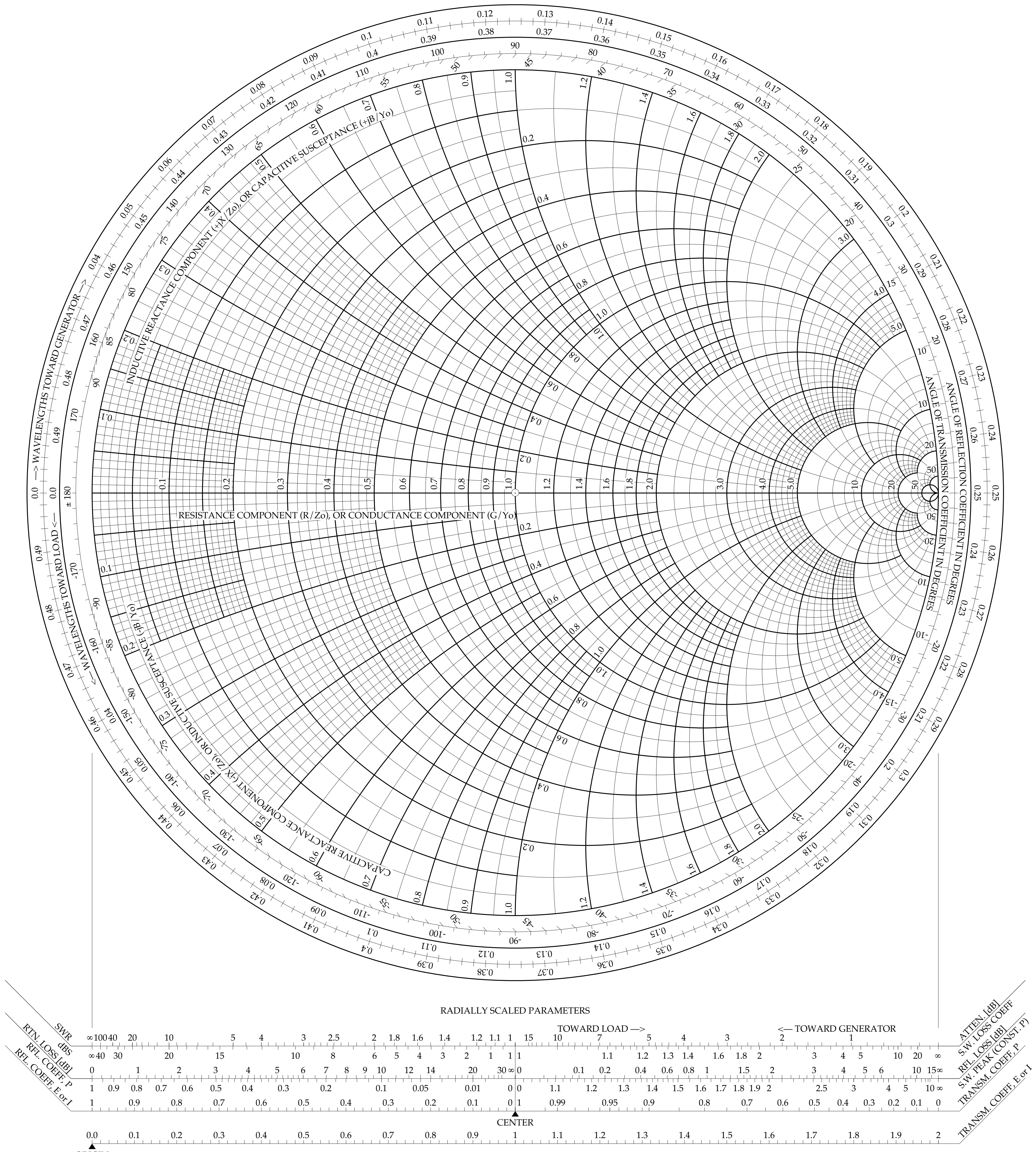

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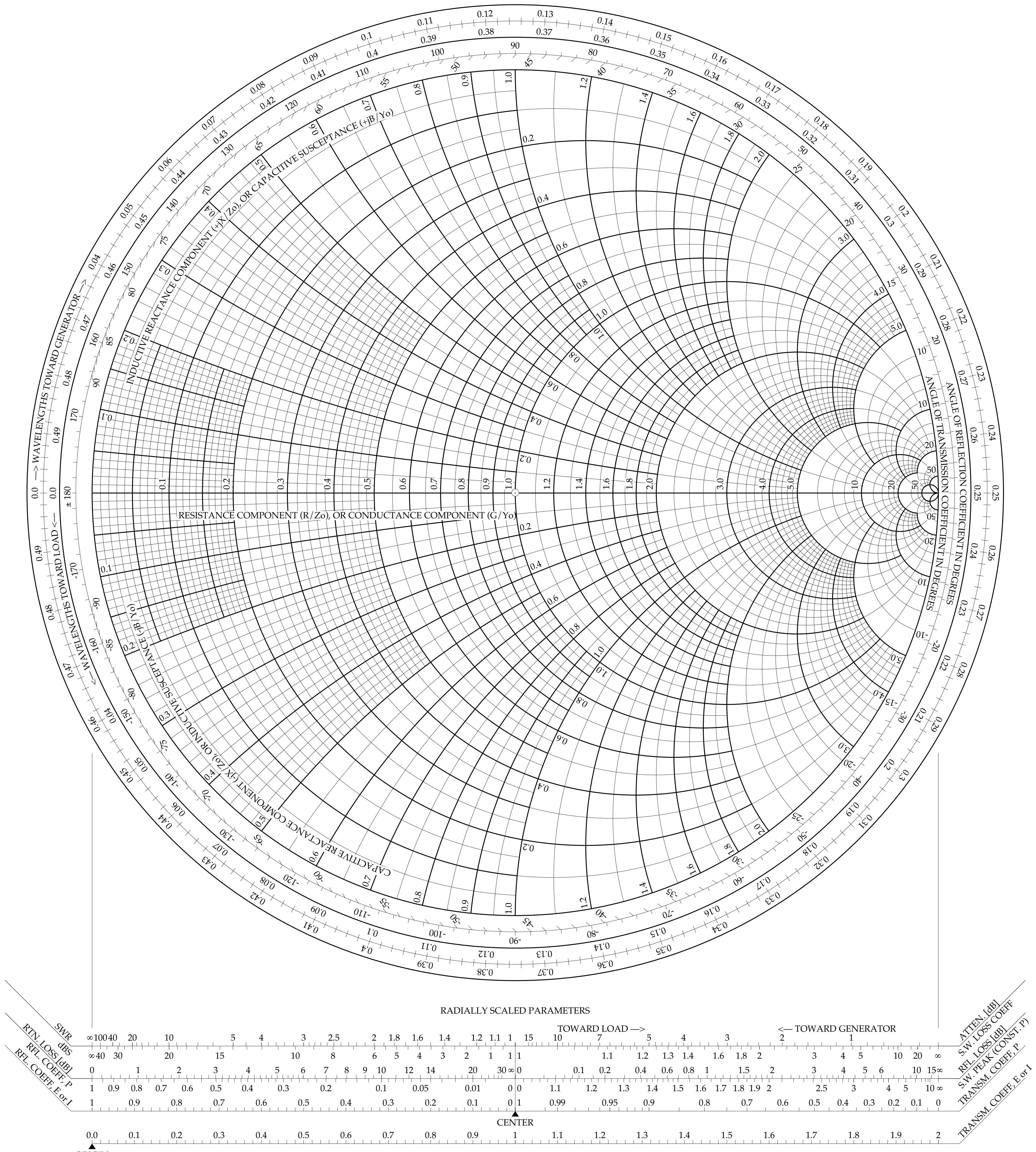

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